Rohan A M

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Profile Summary

Generative AI Developer skilled in **LangChain, RAG, LLMs , Agents & Agentic AI** — building real-world AI tools with startup-level ownership and execution.

Experience (Self-Initiated Projects)

AI and Gen-AI Developer
Apr 2024 - Present (Self Driven)

- **JackUP (AI Powered Research Tool) : Designed and built a fully local AI Research tool for Podcast creators to Streamline their Content creation Process.**
  - Developed a modular **multi-agent system** automating key stages of podcast pre-production using open-source LLM's Like **Mistral** and **Llama3** via **Ollama.**
  - Created an **Intuitive System** reducing research time by **20%**, enabling creators to focus on creativity rather than prep-work.
  - Planning to Integrate **Retrieval-Augmented Generation(RAG)** to enhance contextual research and deliver smarter, more accurate topic recommendations.
  - Envisioning the next phase as a **real-time co-pilot** offering guidance and smart recommendations.
  - **Stack: Ollama, LLama3, Python, Mistral, Multi-Agent System, CLI, Local LLM's, RAG.**

- **Wavely (LinkedIn Post Analysis Application): Built an Electron.js dekstop Overlay App that assists users in writing high-engagement linkedIn posts with real-time LLM Analysis.**
  - The application integrates **Deepseek-coder** for coding related posts and **Gemma3-4B-Instruct** for post analysis and follow-up questions.
  - Integrated **in-app authentication** with **bcrypt-secure** passwords and local session handling.
  - Implemented a realtime **chat-interface** and **structured streaming LLM Output.**(Headings, bullet-points, paragraphs)
  - Engineered **hotkey-driven UX** for minimal disruption during user workflows.
  - **Stack: Electron.js, Node.js, TailwindCSS, DeepseekCoder, Gemma3, LocalAPI's. LMStudio**

- **Hush (Noise Cancellation System Implementation) : Open source noise Cancellation System simulating real-world ANC (Active Noise Cancellation) pipeline.**
  - Implemented **LMS Filtering** to compare Clean v/s Noisy v/s Denoised audio in real-time.
  - Built an Interactive **StreamLit Dashboard** with **matplotlib visualization** for waveform analysis and audio playback.
  - Integrated **UrbanSound8K** dataset to Inject realworld into clean audio samples for realistic testing.
  - Bridged **AI and DSP(Digital signal processing)** by replicating the pipeline AI Audio devices use.
  - **Stack: Python, StreamLit, MatplotLib, NumPy, UrbanSound8K, Signal Processing.**

Education

Bangalore Institute of Technology (BIT) Bangalore, India.
B.Tech in CSE |2020– 2024
CGPA : 7.63 / 10

Skills

Languages: Python, C++
AI/ML: TensorFlow, NLP, Machine Learning, Deep Learning, RAG(Adaptive/Corrective/Agentic/Traditional), LLM Fine-Tuning, Agentic Systems
Frameworks & Tools: LangChain, CrewAI, Streamlit, OpenAI, Groq, AWS(Bedrock,Lambda,S3), STT/TTS APIs,Hugging Face,Langgraph,Replit,Cursor AI, n8n workflow.
Databases: FAISS(Vector DB), Pinecone(Hybrid DB), Neo4j(GraphDB), Supabase.